Bilderwand – layout candidates

Target 180x120 cm ± 10.0 cm, nominal gap free in [1.5, 6.0] cm, orientation fixed, sizes are measured outer dimensions. Max. gap deviation = largest difference between an actual gap and the layout's nominal gap; notches = parts of the outer border not covered by a frame or a regular gap. Ranked by a combined score (gap uniformity first, border sharpness second, then number of frames).

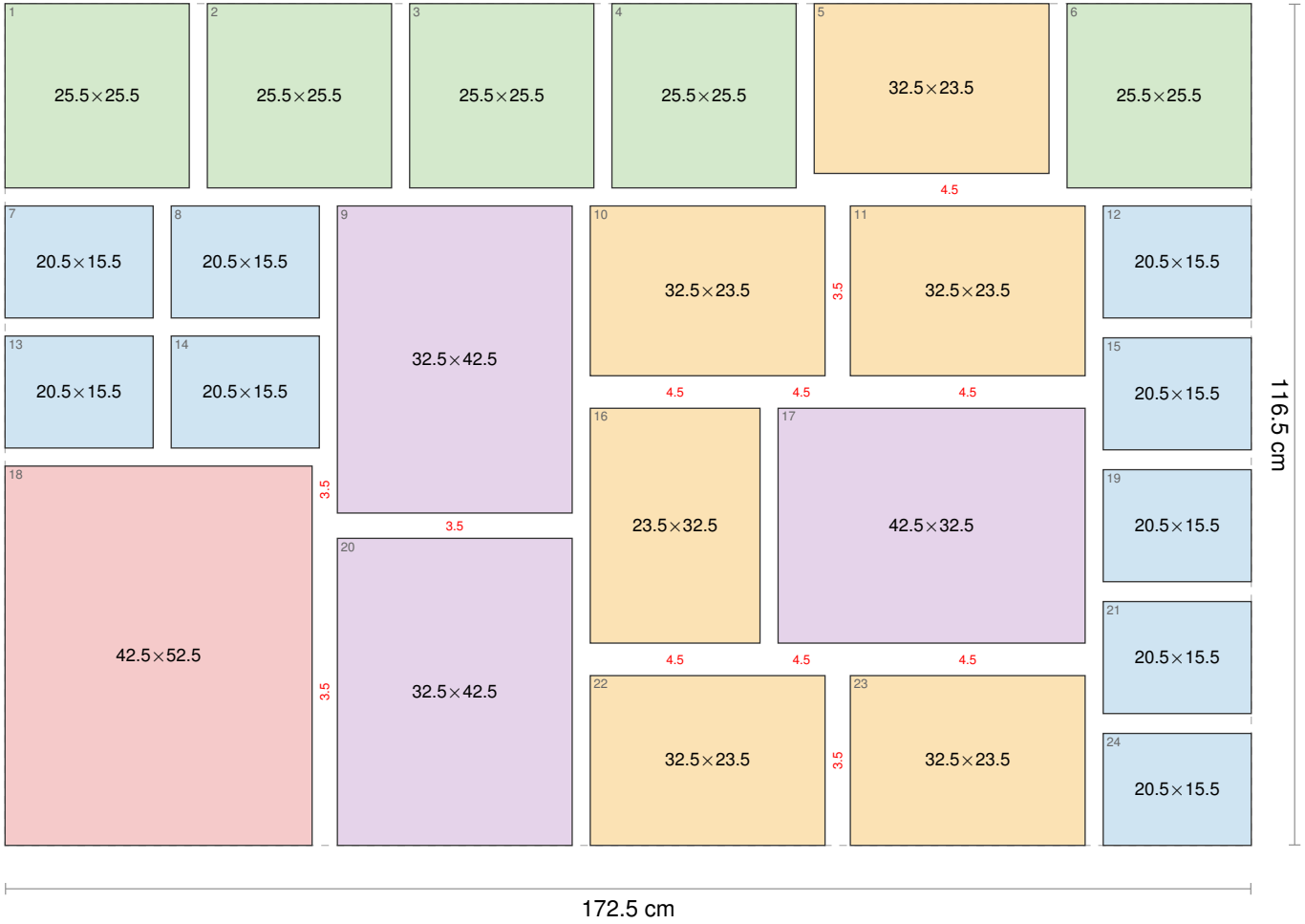
#	size (cm)	gap	frames	max dev	mean dev	notches	corners	entropy
1	172.5 × 116.5	2.5	24	2.0	0.43	none	4/4	0.72
2	181.5 × 113.5	1.5	20	2.0	0.62	none	4/4	0.78
3	187.5 × 113.5	1.5	25	2.5	0.51	4 cm	4/4	0.80
4	187.5 × 113.5	1.5	25	2.5	0.54	4 cm	4/4	0.79
5	177.5 × 126.5	2.5	25	3.0	0.79	none	4/4	0.79
6	172.5 × 124.5	2.5	23	2.5	0.80	none	4/4	0.75
7	176.0 × 123.0	2	23	3.0	0.45	none	4/4	0.82
8	184.0 × 115.0	2	23	2.0	0.54	6 cm	4/4	0.76
9	177.5 × 113.5	1.5	21	2.0	0.56	6 cm	4/4	0.78
10	170.5 × 115.5	2	21	3.0	0.80	none	4/4	0.79
11	170.5 × 117.5	2.5	19	3.0	0.67	none	4/4	0.84
12	179.5 × 120.5	1.5	26	2.5	0.55	10 cm	4/4	0.80
13	179.5 × 120.5	1.5	26	3.0	0.56	7 cm	4/4	0.79
14	172.5 × 116.5	2.5	25	3.0	0.53	4 cm	4/4	0.78
15	177.0 × 123.0	2	25	2.0	0.64	12 cm	4/4	0.79
16	187.5 × 125.0	2	25	3.8	0.86	none	4/4	0.78

Inventory

	outer w × h (cm)		have
	20.5×15.5	L	9
	15.5×20.5	P	2
	32.5×42.5	P	2
	42.5×32.5	L	1
	32.5×23.5	L	5
	23.5×32.5	P	1
	25.5×25.5		5
	42.5×52.5	P	1
	52.5×42.5	L	2

Search parameters: target = 180x120, tol = 10.0, gap_min = 1.5, gap_max = 6.0, gap_mode = free, delta = 0.5, slack_rule = even, cost_shape = huber, min_frames = 10, frame_bonus = 1.5, runs = 256, iters = 20 min, seed = 0, entropy = 40.0

Layout 1 172.5 × 116.5 cm gap 2.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

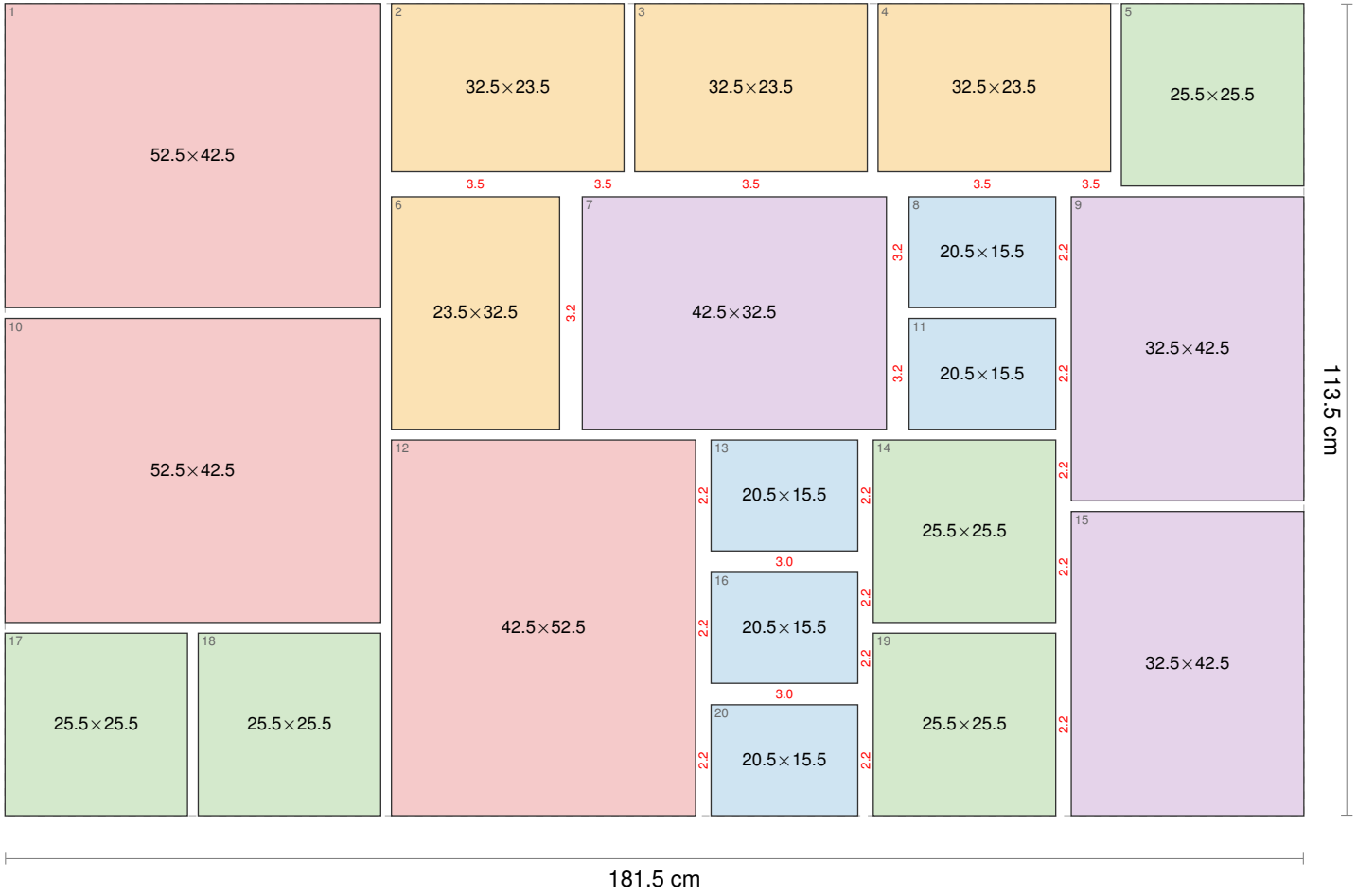
Outer size	172.5 × 116.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 4.5 cm
Max. deviation	2.0 cm
Mean deviation	0.43 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	1
Entropy	0.72
mix / variety	0.74 / 0.65
stagger / lines	0.49 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	25.5 × 25.5	28.0	0.0
3	25.5 × 25.5	56.0	0.0
4	25.5 × 25.5	84.0	0.0
5	32.5 × 23.5	112.0	0.0
6	25.5 × 25.5	147.0	0.0
7	20.5 × 15.5	0.0	28.0
8	20.5 × 15.5	23.0	28.0
9	32.5 × 42.5	46.0	28.0
10	32.5 × 23.5	81.0	28.0
11	32.5 × 23.5	117.0	28.0
12	20.5 × 15.5	152.0	28.0
13	20.5 × 15.5	0.0	46.0
14	20.5 × 15.5	23.0	46.0
15	20.5 × 15.5	152.0	46.2
16	23.5 × 32.5	81.0	56.0
17	42.5 × 32.5	107.0	56.0
18	42.5 × 52.5	0.0	64.0
19	20.5 × 15.5	152.0	64.5
20	32.5 × 42.5	46.0	74.0
21	20.5 × 15.5	152.0	82.8
22	32.5 × 23.5	81.0	93.0
23	32.5 × 23.5	117.0	93.0
24	20.5 × 15.5	152.0	101.0

Layout 2 181.5 × 113.5 cm gap 1.5 cm 20 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

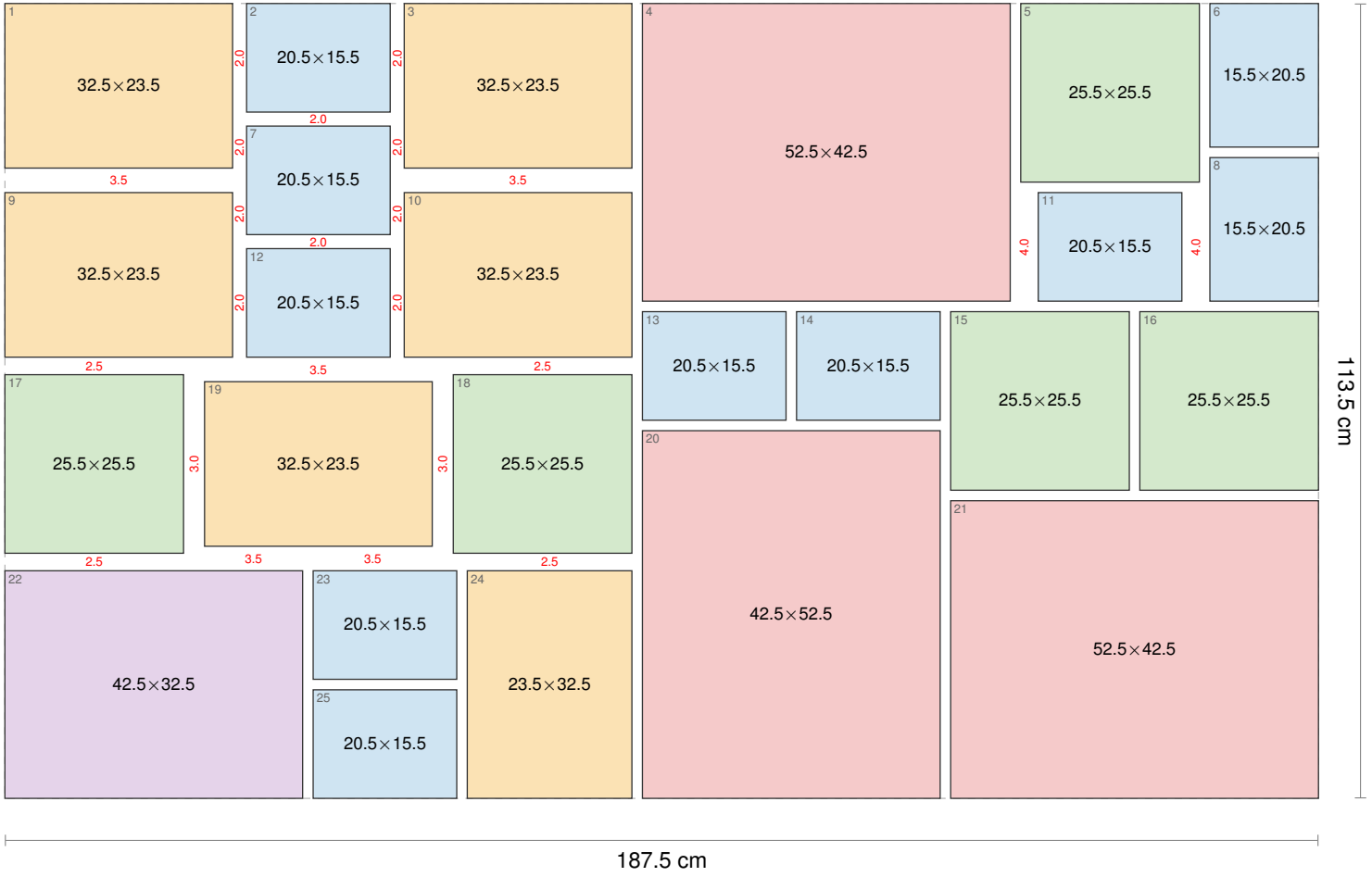
Outer size	181.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 3.5 cm
Max. deviation	2.0 cm
Mean deviation	0.62 cm
Border notches	none
Corners filled	4 / 4
Frames used	20 of 28
Large frames	3, touching 2× (78 cm), min. dist. 44 cm
Entropy	0.78
mix / variety	0.86 / 0.78
stagger / lines	0.49 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	5	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 23.5	54.0	0.0
3	32.5 × 23.5	88.0	0.0
4	32.5 × 23.5	122.0	0.0
5	25.5 × 25.5	156.0	0.0
6	23.5 × 32.5	54.0	27.0
7	42.5 × 32.5	80.7	27.0
8	20.5 × 15.5	126.3	27.0
9	32.5 × 42.5	149.0	27.0
10	52.5 × 42.5	0.0	44.0
11	20.5 × 15.5	126.3	44.0
12	42.5 × 52.5	54.0	61.0
13	20.5 × 15.5	98.7	61.0
14	25.5 × 25.5	121.3	61.0
15	32.5 × 42.5	149.0	71.0
16	20.5 × 15.5	98.7	79.5
17	25.5 × 25.5	0.0	88.0
18	25.5 × 25.5	27.0	88.0
19	25.5 × 25.5	121.3	88.0
20	20.5 × 15.5	98.7	98.0

Layout 3 187.5 × 113.5 cm gap 1.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

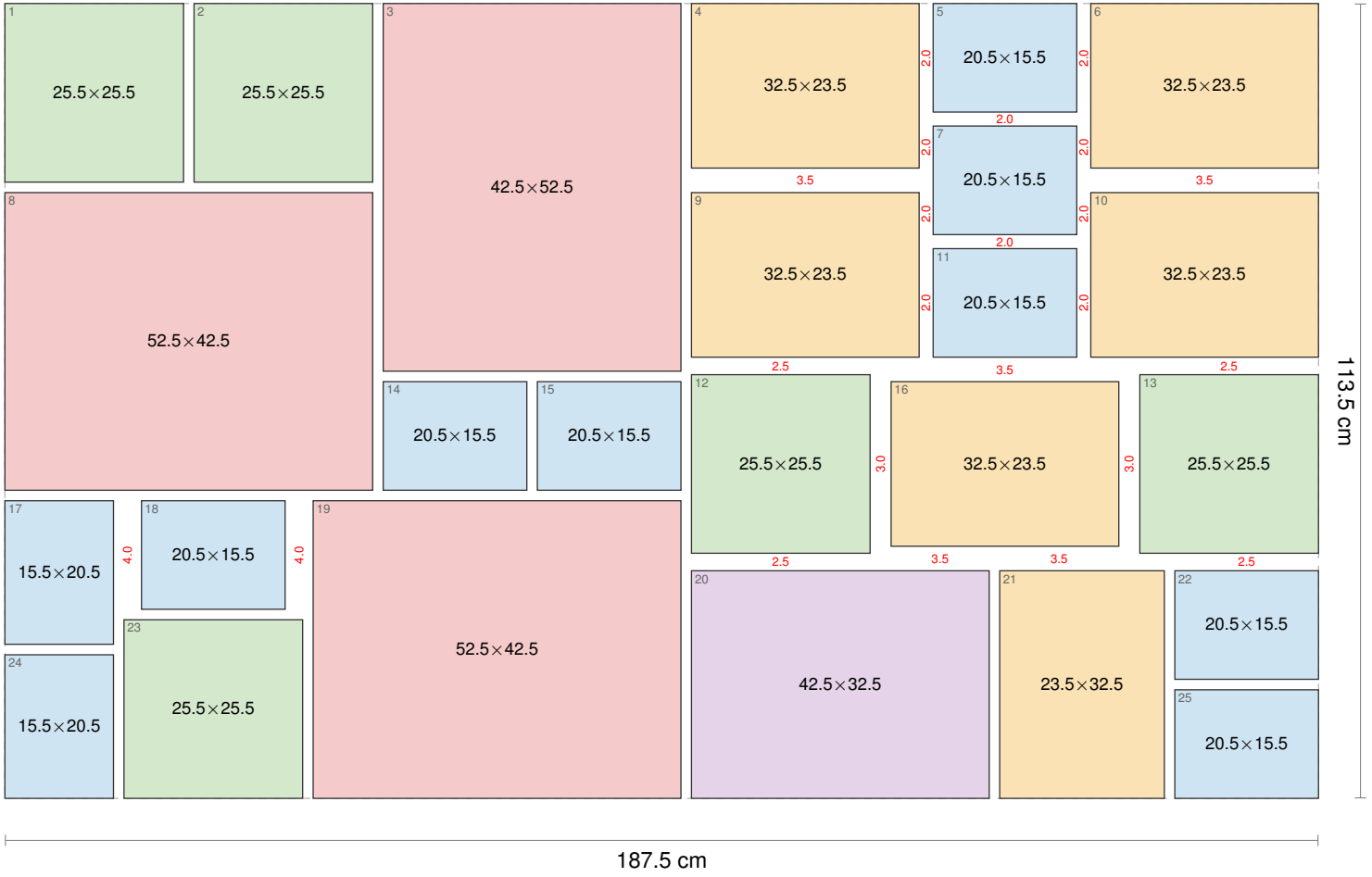
Outer size	187.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.0 cm
Max. deviation	2.5 cm
Mean deviation	0.51 cm
Border notches	left 4 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	3, touching 1× (42 cm), min. dist. 49 cm
Entropy	0.80
mix / variety	0.82 / 0.86
stagger / lines	0.53 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	2	2
	32.5×42.5	P	0	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	20.5 × 15.5	34.5	0.0
3	32.5 × 23.5	57.0	0.0
4	52.5 × 42.5	91.0	0.0
5	25.5 × 25.5	145.0	0.0
6	15.5 × 20.5	172.0	0.0
7	20.5 × 15.5	34.5	17.5
8	15.5 × 20.5	172.0	22.0
9	32.5 × 23.5	0.0	27.0
10	32.5 × 23.5	57.0	27.0
11	20.5 × 15.5	147.5	27.0
12	20.5 × 15.5	34.5	35.0
13	20.5 × 15.5	91.0	44.0
14	20.5 × 15.5	113.0	44.0
15	25.5 × 25.5	135.0	44.0
16	25.5 × 25.5	162.0	44.0
17	25.5 × 25.5	0.0	53.0
18	25.5 × 25.5	64.0	53.0
19	32.5 × 23.5	28.5	54.0
20	42.5 × 52.5	91.0	61.0
21	52.5 × 42.5	135.0	71.0
22	42.5 × 32.5	0.0	81.0
23	20.5 × 15.5	44.0	81.0
24	23.5 × 32.5	66.0	81.0
25	20.5 × 15.5	44.0	98.0

Layout 4 187.5 × 113.5 cm gap 1.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	187.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.0 cm
Max. deviation	2.5 cm
Mean deviation	0.54 cm
Border notches	right 4 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	3, touching 2× (34 cm), min. dist. 54 cm
Entropy	0.79
mix / variety	0.82 / 0.83
stagger / lines	0.52 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	2	2
	32.5×42.5	P	0	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	25.5 × 25.5	27.0	0.0
3	42.5 × 52.5	54.0	0.0
4	32.5 × 23.5	98.0	0.0
5	20.5 × 15.5	132.5	0.0
6	32.5 × 23.5	155.0	0.0
7	20.5 × 15.5	132.5	17.5
8	52.5 × 42.5	0.0	27.0
9	32.5 × 23.5	98.0	27.0
10	32.5 × 23.5	155.0	27.0
11	20.5 × 15.5	132.5	35.0
12	25.5 × 25.5	98.0	53.0
13	25.5 × 25.5	162.0	53.0
14	20.5 × 15.5	54.0	54.0
15	20.5 × 15.5	76.0	54.0
16	32.5 × 23.5	126.5	54.0
17	15.5 × 20.5	0.0	71.0
18	20.5 × 15.5	19.5	71.0
19	52.5 × 42.5	44.0	71.0
20	42.5 × 32.5	98.0	81.0
21	23.5 × 32.5	142.0	81.0
22	20.5 × 15.5	167.0	81.0
23	25.5 × 25.5	17.0	88.0
24	15.5 × 20.5	0.0	93.0
25	20.5 × 15.5	167.0	98.0

Layout 5 177.5 × 126.5 cm gap 2.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

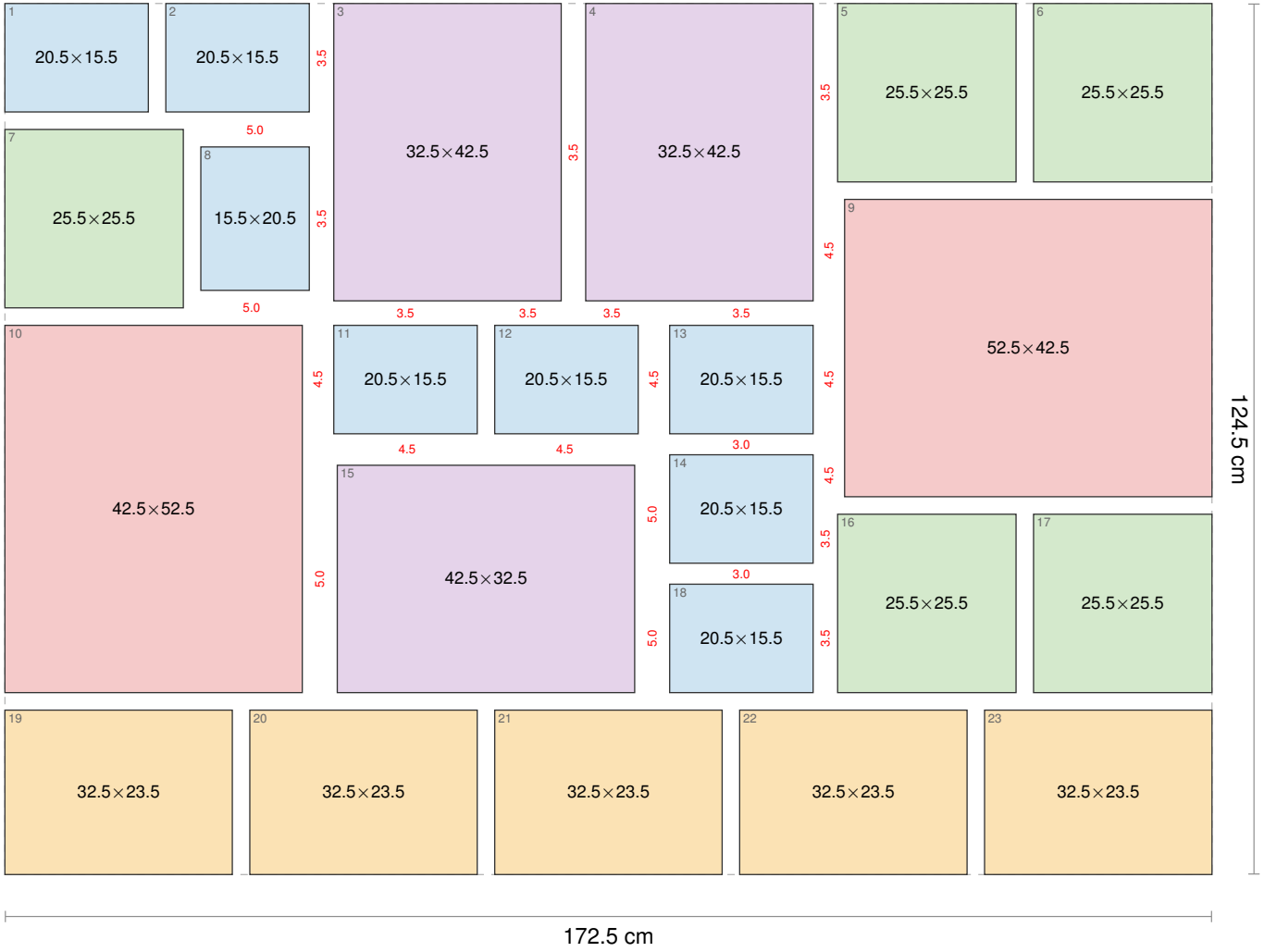
Outer size	177.5 × 126.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 5.5 cm
Max. deviation	3.0 cm
Mean deviation	0.79 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 90 cm
Entropy	0.79
mix / variety	0.81 / 0.81
stagger / lines	0.56 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	20.5 × 15.5	23.7	0.0
3	20.5 × 15.5	47.3	0.0
4	20.5 × 15.5	71.0	0.0
5	32.5 × 23.5	94.0	0.0
6	20.5 × 15.5	129.0	0.0
7	25.5 × 25.5	152.0	0.0
8	25.5 × 25.5	0.0	19.0
9	25.5 × 25.5	66.0	19.0
10	32.5 × 23.5	29.5	20.0
11	20.5 × 15.5	129.0	20.3
12	25.5 × 25.5	152.0	28.0
13	32.5 × 23.5	94.0	29.0
14	20.5 × 15.5	129.0	40.7
15	20.5 × 15.5	0.0	48.0
16	42.5 × 32.5	23.0	48.5
17	23.5 × 32.5	68.0	48.5
18	25.5 × 25.5	152.0	56.0
19	32.5 × 23.5	94.0	58.0
20	15.5 × 20.5	131.5	61.0
21	20.5 × 15.5	0.0	66.0
22	32.5 × 42.5	0.0	84.0
23	52.5 × 42.5	35.0	84.0
24	32.5 × 42.5	90.0	84.0
25	52.5 × 42.5	125.0	84.0

Layout 6 172.5 × 124.5 cm gap 2.5 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

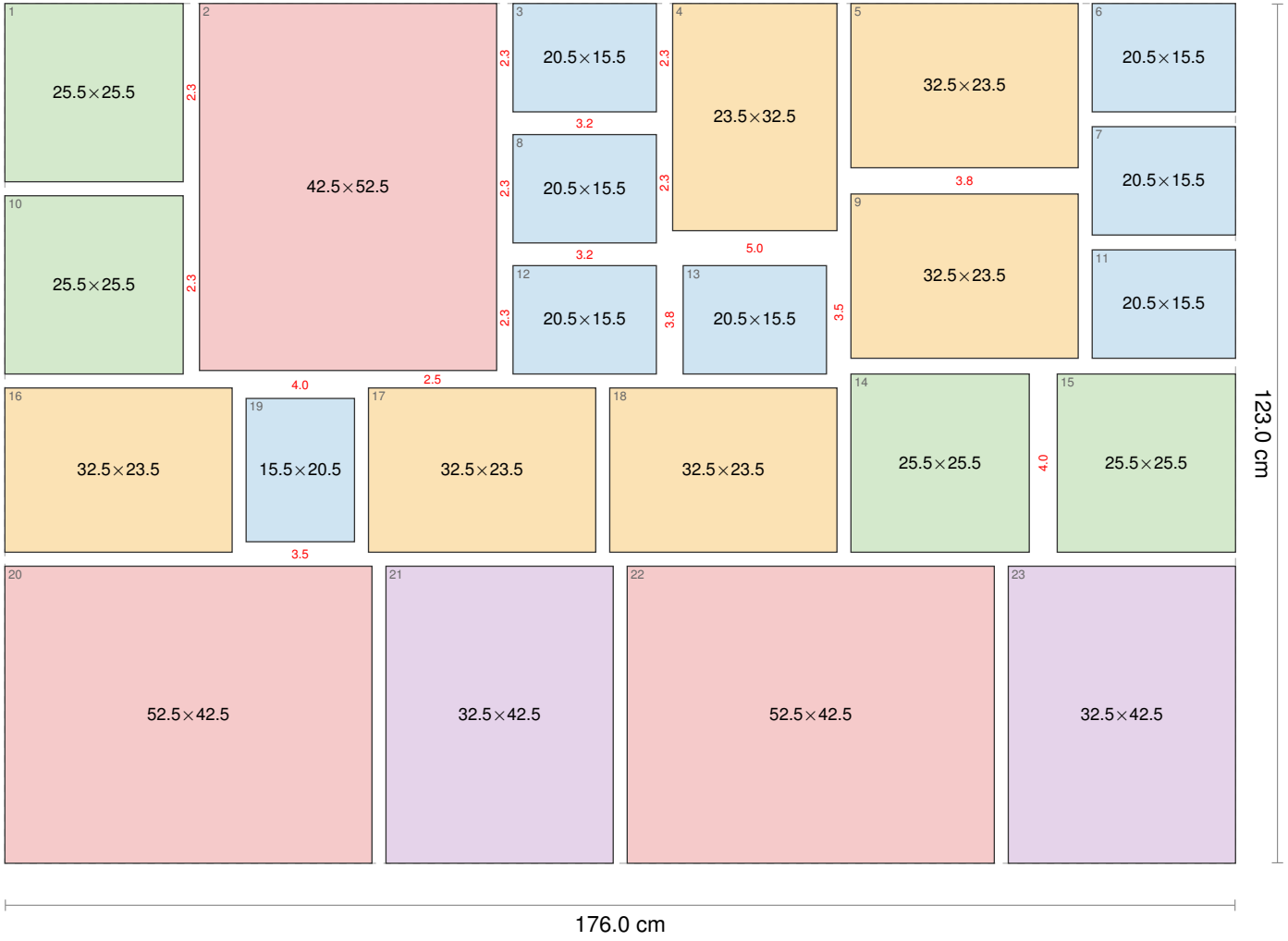
Outer size	172.5 × 124.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 5.0 cm
Max. deviation	2.5 cm
Mean deviation	0.80 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	2, not touching, min. dist. 127 cm
Entropy	0.75
mix / variety	0.81 / 0.69
stagger / lines	0.52 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	7	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	0	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	20.5 × 15.5	23.0	0.0
3	32.5 × 42.5	47.0	0.0
4	32.5 × 42.5	83.0	0.0
5	25.5 × 25.5	119.0	0.0
6	25.5 × 25.5	147.0	0.0
7	25.5 × 25.5	0.0	18.0
8	15.5 × 20.5	28.0	20.5
9	52.5 × 42.5	120.0	28.0
10	42.5 × 52.5	0.0	46.0
11	20.5 × 15.5	47.0	46.0
12	20.5 × 15.5	70.0	46.0
13	20.5 × 15.5	95.0	46.0
14	20.5 × 15.5	95.0	64.5
15	42.5 × 32.5	47.5	66.0
16	25.5 × 25.5	119.0	73.0
17	25.5 × 25.5	147.0	73.0
18	20.5 × 15.5	95.0	83.0
19	32.5 × 23.5	0.0	101.0
20	32.5 × 23.5	35.0	101.0
21	32.5 × 23.5	70.0	101.0
22	32.5 × 23.5	105.0	101.0
23	32.5 × 23.5	140.0	101.0

Layout 7 176.0 × 123.0 cm gap 2 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	176.0 × 123.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 5.0 cm
Max. deviation	3.0 cm
Mean deviation	0.45 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, not touching, min. dist. 79 cm
Entropy	0.82
mix / variety	0.83 / 0.89
stagger / lines	0.54 / 1.00 (1 through)

outer w × h (cm)		used	have
20.5×15.5	L	7	9
15.5×20.5	P	1	2
32.5×42.5	P	2	2
42.5×32.5	L	0	1
32.5×23.5	L	5	5
23.5×32.5	P	1	1
25.5×25.5		4	5
42.5×52.5	P	1	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	42.5 × 52.5	27.8	0.0
3	20.5 × 15.5	72.7	0.0
4	23.5 × 32.5	95.5	0.0
5	32.5 × 23.5	121.0	0.0
6	20.5 × 15.5	155.5	0.0
7	20.5 × 15.5	155.5	17.6
8	20.5 × 15.5	72.7	18.8
9	32.5 × 23.5	121.0	27.2
10	25.5 × 25.5	0.0	27.5
11	20.5 × 15.5	155.5	35.2
12	20.5 × 15.5	72.7	37.5
13	20.5 × 15.5	97.0	37.5
14	25.5 × 25.5	121.0	53.0
15	25.5 × 25.5	150.5	53.0
16	32.5 × 23.5	0.0	55.0
17	32.5 × 23.5	52.0	55.0
18	32.5 × 23.5	86.5	55.0
19	15.5 × 20.5	34.5	56.5
20	52.5 × 42.5	0.0	80.5
21	32.5 × 42.5	54.5	80.5
22	52.5 × 42.5	89.0	80.5
23	32.5 × 42.5	143.5	80.5

Layout 8 184.0 × 115.0 cm gap 2 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

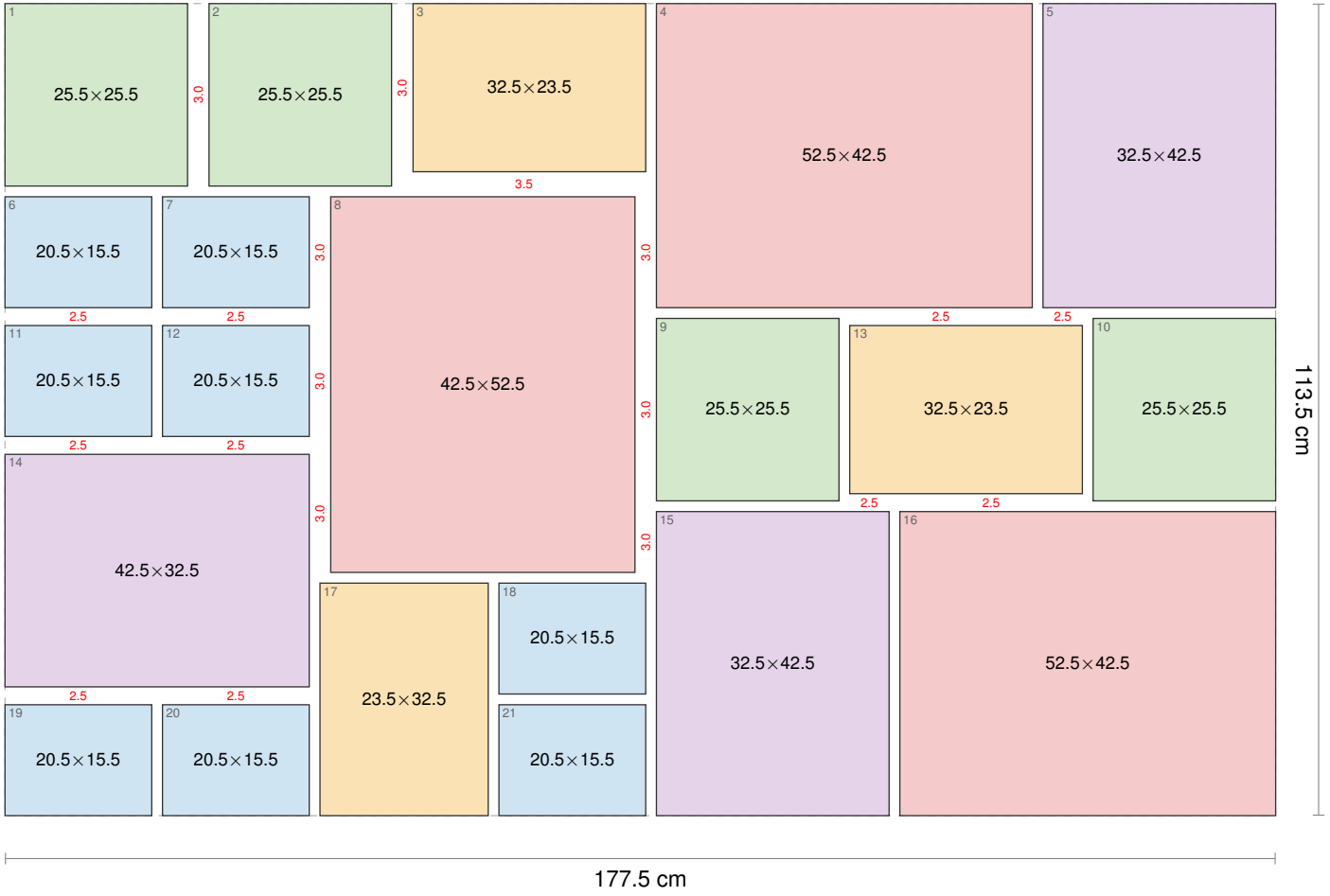
Outer size	184.0 × 115.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.0 cm
Max. deviation	2.0 cm
Mean deviation	0.54 cm
Border notches	right 6 cm
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, touching 1× (52 cm), min. dist. 45 cm
Entropy	0.76
mix / variety	0.76 / 0.76
stagger / lines	0.54 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 23.5	27.5	0.0
3	25.5 × 25.5	62.0	0.0
4	32.5 × 23.5	89.5	0.0
5	25.5 × 25.5	124.0	0.0
6	32.5 × 42.5	151.5	0.0
7	52.5 × 42.5	0.0	27.5
8	20.5 × 15.5	55.2	27.5
9	23.5 × 32.5	79.3	27.5
10	42.5 × 52.5	106.8	27.5
11	20.5 × 15.5	55.2	45.5
12	32.5 × 42.5	151.5	45.8
13	25.5 × 25.5	78.3	62.0
14	20.5 × 15.5	55.2	63.5
15	52.5 × 42.5	0.0	72.5
16	20.5 × 15.5	55.2	81.5
17	20.5 × 15.5	106.5	82.0
18	20.5 × 15.5	129.0	82.0
19	25.5 × 25.5	78.3	89.5
20	32.5 × 23.5	151.5	91.5
21	20.5 × 15.5	55.2	99.5
22	20.5 × 15.5	106.5	99.5
23	20.5 × 15.5	129.0	99.5

Layout 9 177.5 × 113.5 cm gap 1.5 cm 21 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

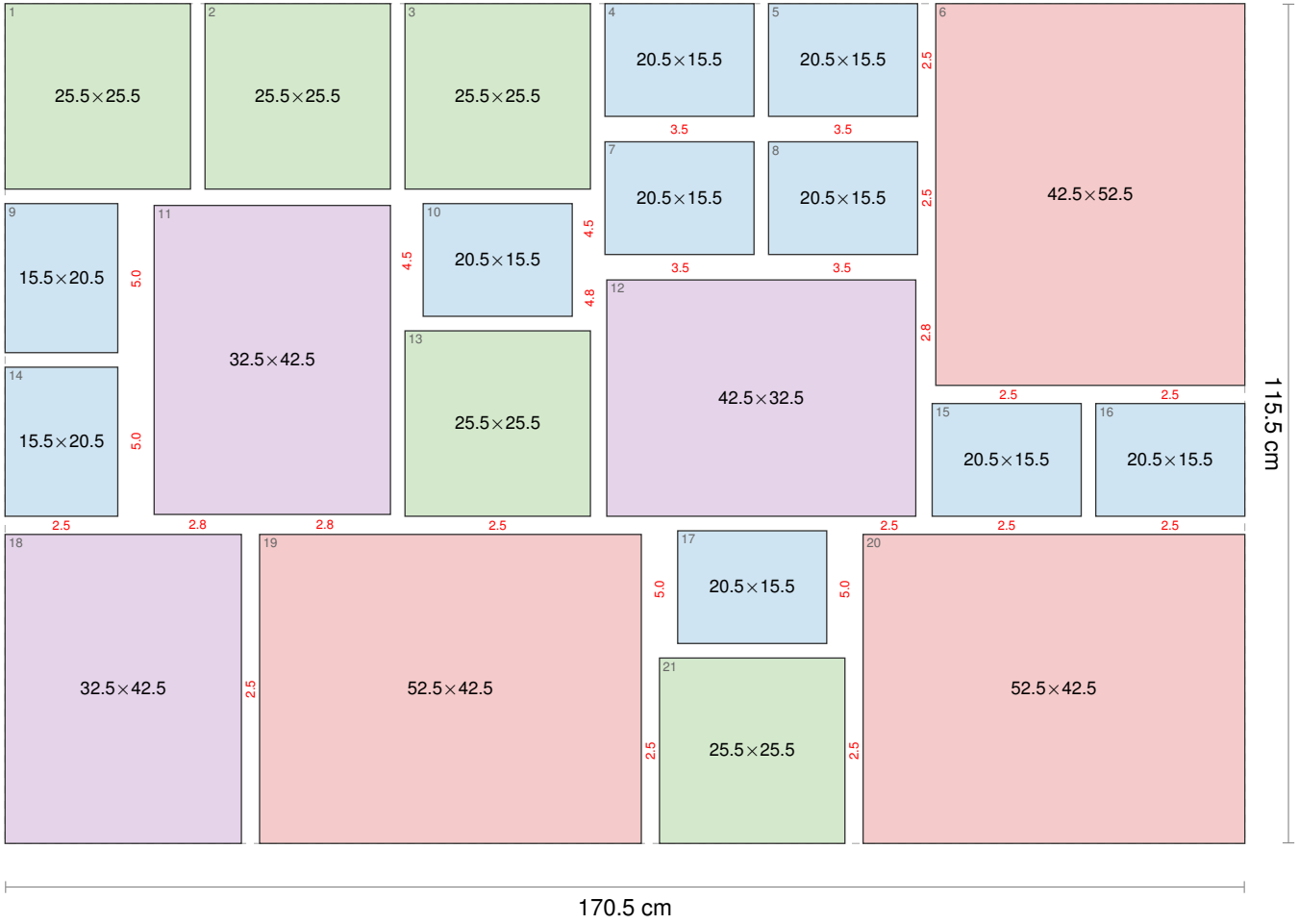
Outer size	177.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 3.5 cm
Max. deviation	2.0 cm
Mean deviation	0.56 cm
Border notches	top 6 cm
Corners filled	4 / 4
Frames used	21 of 28
Large frames	3, not touching, min. dist. 60 cm
Entropy	0.78
mix / variety	0.81 / 0.82
stagger / lines	0.50 / 1.00 (1 through)

outer w × h (cm)		used	have
20.5×15.5	L	8	9
15.5×20.5	P	0	2
32.5×42.5	P	2	2
42.5×32.5	L	1	1
32.5×23.5	L	2	5
23.5×32.5	P	1	1
25.5×25.5		4	5
42.5×52.5	P	1	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	25.5 × 25.5	28.5	0.0
3	32.5 × 23.5	57.0	0.0
4	52.5 × 42.5	91.0	0.0
5	32.5 × 42.5	145.0	0.0
6	20.5 × 15.5	0.0	27.0
7	20.5 × 15.5	22.0	27.0
8	42.5 × 52.5	45.5	27.0
9	25.5 × 25.5	91.0	44.0
10	25.5 × 25.5	152.0	44.0
11	20.5 × 15.5	0.0	45.0
12	20.5 × 15.5	22.0	45.0
13	32.5 × 23.5	118.0	45.0
14	42.5 × 32.5	0.0	63.0
15	32.5 × 42.5	91.0	71.0
16	52.5 × 42.5	125.0	71.0
17	23.5 × 32.5	44.0	81.0
18	20.5 × 15.5	69.0	81.0
19	20.5 × 15.5	0.0	98.0
20	20.5 × 15.5	22.0	98.0
21	20.5 × 15.5	69.0	98.0

Layout 10 170.5 × 115.5 cm gap 2 cm 21 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	170.5 × 115.5 cm
Nominal gap	2 cm
Actual gaps	2.0 – 5.0 cm
Max. deviation	3.0 cm
Mean deviation	0.80 cm
Border notches	none
Corners filled	4 / 4
Frames used	21 of 28
Large frames	3, not touching, min. dist. 68 cm
Entropy	0.79
mix / variety	0.76 / 0.79
stagger / lines	0.60 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	0	5
	23.5×32.5	P	0	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	25.5 × 25.5	27.5	0.0
3	25.5 × 25.5	55.0	0.0
4	20.5 × 15.5	82.5	0.0
5	20.5 × 15.5	105.0	0.0
6	42.5 × 52.5	128.0	0.0
7	20.5 × 15.5	82.5	19.0
8	20.5 × 15.5	105.0	19.0
9	15.5 × 20.5	0.0	27.5
10	20.5 × 15.5	57.5	27.5
11	32.5 × 42.5	20.5	27.8
12	42.5 × 32.5	82.8	38.0
13	25.5 × 25.5	55.0	45.0
14	15.5 × 20.5	0.0	50.0
15	20.5 × 15.5	127.5	55.0
16	20.5 × 15.5	150.0	55.0
17	20.5 × 15.5	92.5	72.5
18	32.5 × 42.5	0.0	73.0
19	52.5 × 42.5	35.0	73.0
20	52.5 × 42.5	118.0	73.0
21	25.5 × 25.5	90.0	90.0

Layout 11 170.5 × 117.5 cm gap 2.5 cm 19 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

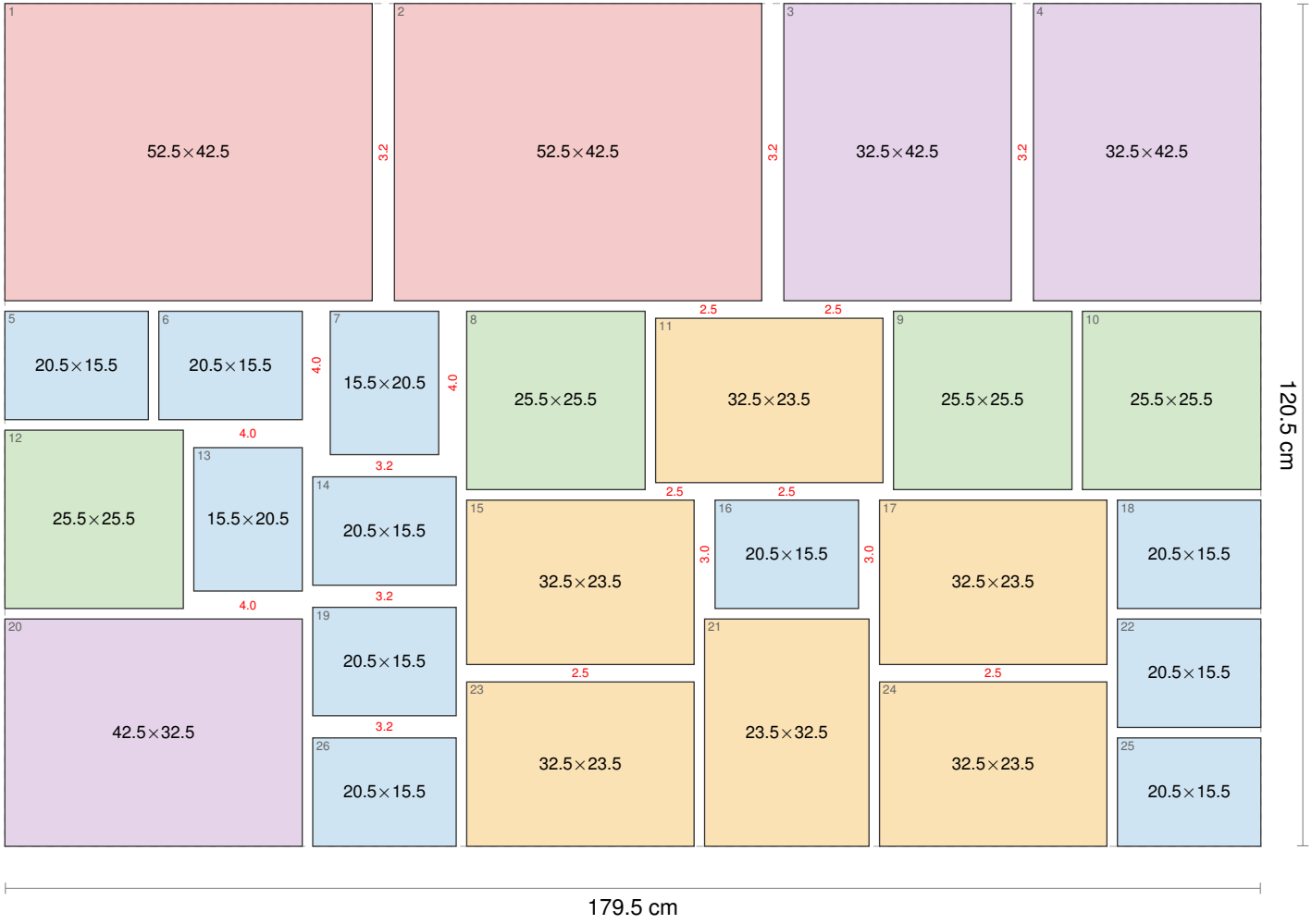
Outer size	170.5 × 117.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 5.5 cm
Max. deviation	3.0 cm
Mean deviation	0.67 cm
Border notches	none
Corners filled	4 / 4
Frames used	19 of 28
Large frames	3, not touching, min. dist. 70 cm
Entropy	0.84
mix / variety	0.89 / 0.86
stagger / lines	0.63 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	3	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	15.5 × 20.5	0.0	0.0
2	15.5 × 20.5	18.0	0.0
3	32.5 × 42.5	36.0	0.0
4	25.5 × 25.5	71.0	0.0
5	42.5 × 52.5	99.5	0.0
6	25.5 × 25.5	145.0	0.0
7	32.5 × 23.5	0.0	23.0
8	20.5 × 15.5	73.5	28.0
9	25.5 × 25.5	145.0	28.3
10	25.5 × 25.5	36.0	47.0
11	32.5 × 23.5	64.0	48.0
12	32.5 × 23.5	0.0	49.0
13	23.5 × 32.5	147.0	56.7
14	20.5 × 15.5	99.0	57.0
15	20.5 × 15.5	122.0	57.0
16	52.5 × 42.5	0.0	75.0
17	32.5 × 42.5	55.0	75.0
18	52.5 × 42.5	90.0	75.0
19	25.5 × 25.5	145.0	92.0

Layout 12 179.5 × 120.5 cm gap 1.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

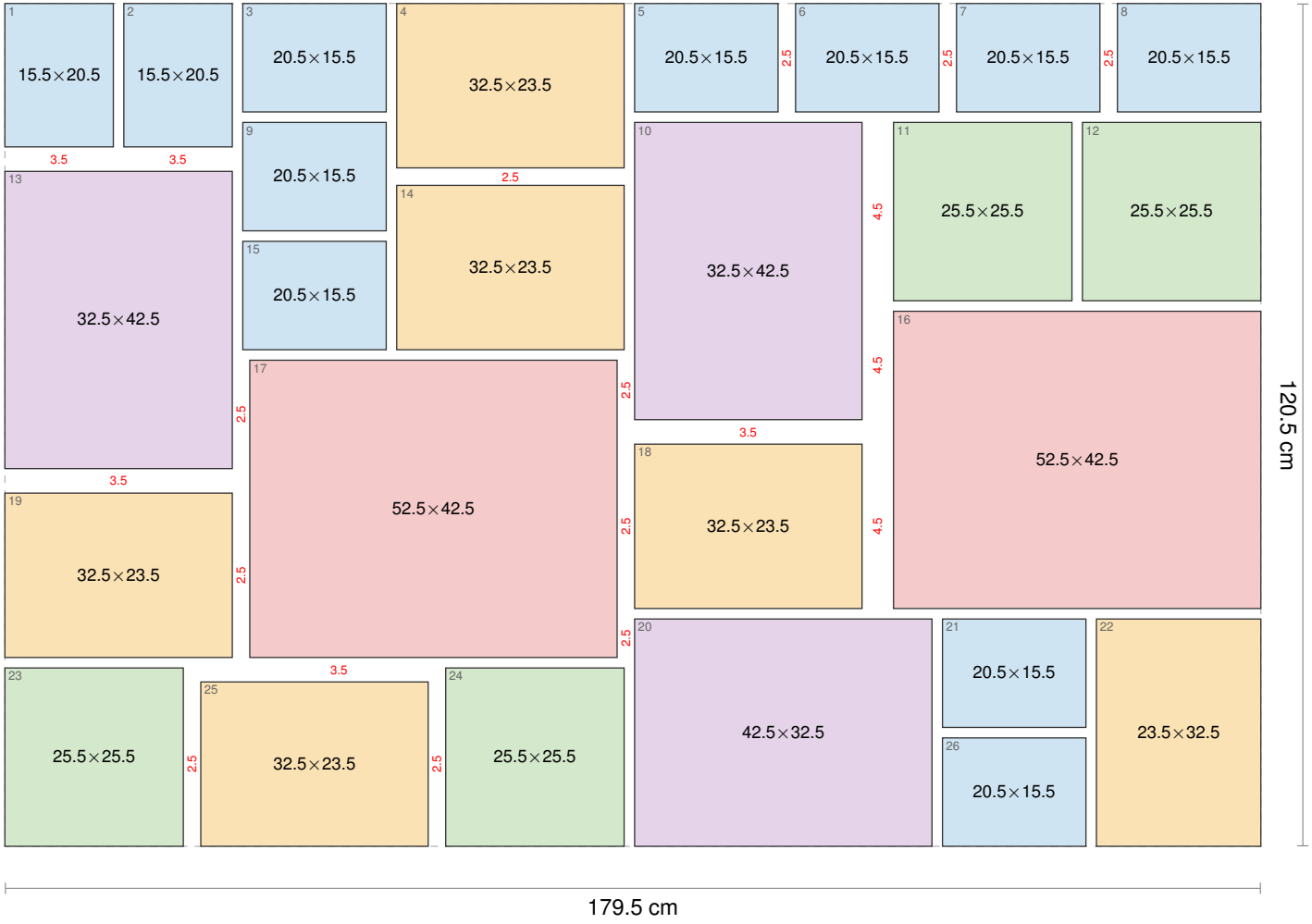
Outer size	179.5 × 120.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.0 cm
Max. deviation	2.5 cm
Mean deviation	0.55 cm
Border notches	top 10 cm
Corners filled	4 / 4
Frames used	26 of 28
Large frames	2, not touching, min. dist. 56 cm
Entropy	0.80
mix / variety	0.83 / 0.83
stagger / lines	0.53 / 1.00 (1 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	2	2
32.5×42.5	P	2	2
42.5×32.5	L	1	1
32.5×23.5	L	5	5
23.5×32.5	P	1	1
25.5×25.5		4	5
42.5×52.5	P	0	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	52.5 × 42.5	55.7	0.0
3	32.5 × 42.5	111.3	0.0
4	32.5 × 42.5	147.0	0.0
5	20.5 × 15.5	0.0	44.0
6	20.5 × 15.5	22.0	44.0
7	15.5 × 20.5	46.5	44.0
8	25.5 × 25.5	66.0	44.0
9	25.5 × 25.5	127.0	44.0
10	25.5 × 25.5	154.0	44.0
11	32.5 × 23.5	93.0	45.0
12	25.5 × 25.5	0.0	61.0
13	15.5 × 20.5	27.0	63.5
14	20.5 × 15.5	44.0	67.7
15	32.5 × 23.5	66.0	71.0
16	20.5 × 15.5	101.5	71.0
17	32.5 × 23.5	125.0	71.0
18	20.5 × 15.5	159.0	71.0
19	20.5 × 15.5	44.0	86.3
20	42.5 × 32.5	0.0	88.0
21	23.5 × 32.5	100.0	88.0
22	20.5 × 15.5	159.0	88.0
23	32.5 × 23.5	66.0	97.0
24	32.5 × 23.5	125.0	97.0
25	20.5 × 15.5	159.0	105.0
26	20.5 × 15.5	44.0	105.0

Layout 13 179.5 × 120.5 cm gap 1.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

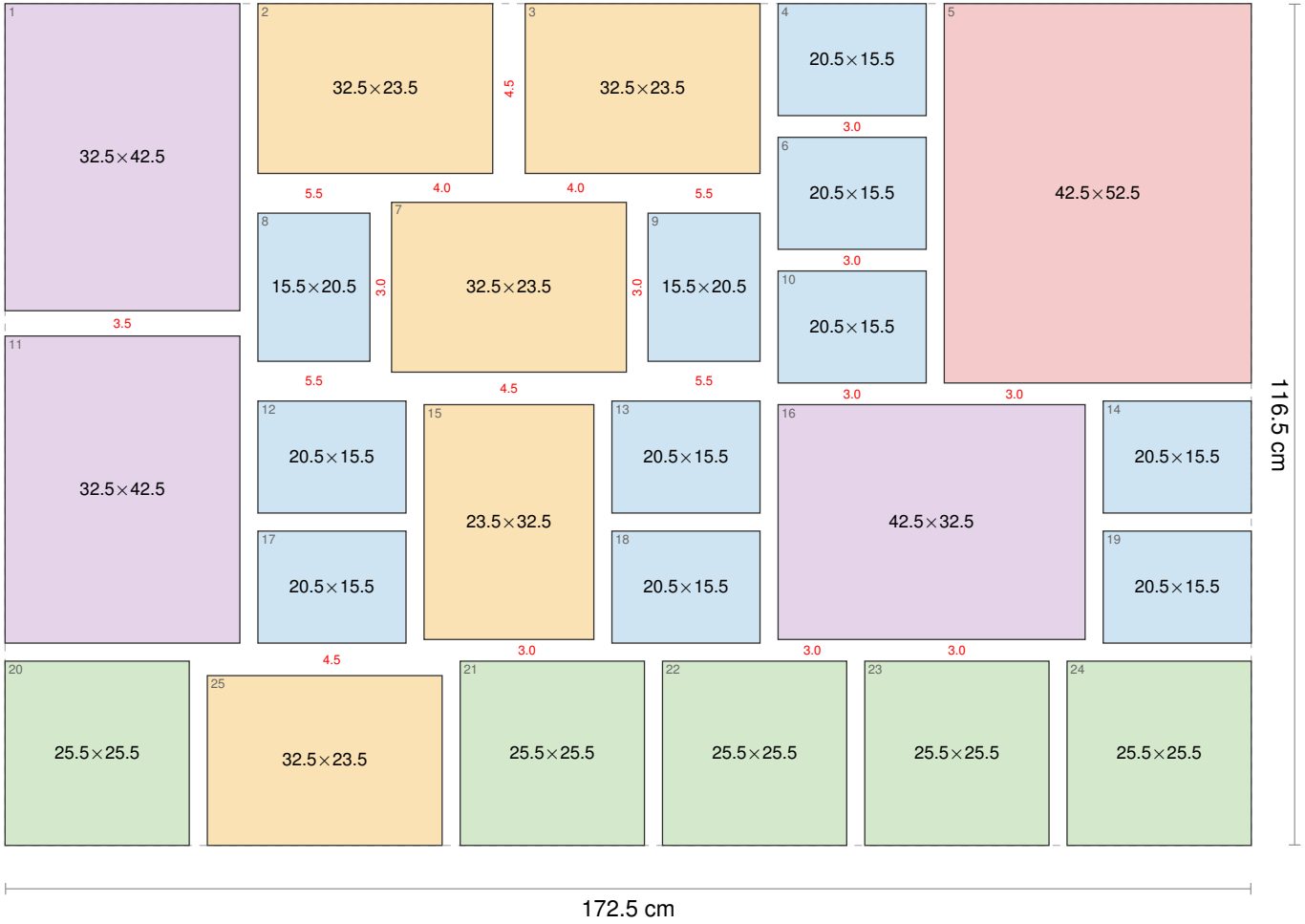
Outer size	179.5 × 120.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.5 cm
Max. deviation	3.0 cm
Mean deviation	0.56 cm
Border notches	left 7 cm
Corners filled	4 / 4
Frames used	26 of 28
Large frames	2, not touching, min. dist. 92 cm
Entropy	0.79
mix / variety	0.83 / 0.80
stagger / lines	0.53 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	15.5 × 20.5	0.0	0.0
2	15.5 × 20.5	17.0	0.0
3	20.5 × 15.5	34.0	0.0
4	32.5 × 23.5	56.0	0.0
5	20.5 × 15.5	90.0	0.0
6	20.5 × 15.5	113.0	0.0
7	20.5 × 15.5	136.0	0.0
8	20.5 × 15.5	159.0	0.0
9	20.5 × 15.5	34.0	17.0
10	32.5 × 42.5	90.0	17.0
11	25.5 × 25.5	127.0	17.0
12	25.5 × 25.5	154.0	17.0
13	32.5 × 42.5	0.0	24.0
14	32.5 × 23.5	56.0	26.0
15	20.5 × 15.5	34.0	34.0
16	52.5 × 42.5	127.0	44.0
17	52.5 × 42.5	35.0	51.0
18	32.5 × 23.5	90.0	63.0
19	32.5 × 23.5	0.0	70.0
20	42.5 × 32.5	90.0	88.0
21	20.5 × 15.5	134.0	88.0
22	23.5 × 32.5	156.0	88.0
23	25.5 × 25.5	0.0	95.0
24	25.5 × 25.5	63.0	95.0
25	32.5 × 23.5	28.0	97.0
26	20.5 × 15.5	134.0	105.0

Layout 14 172.5 × 116.5 cm gap 2.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

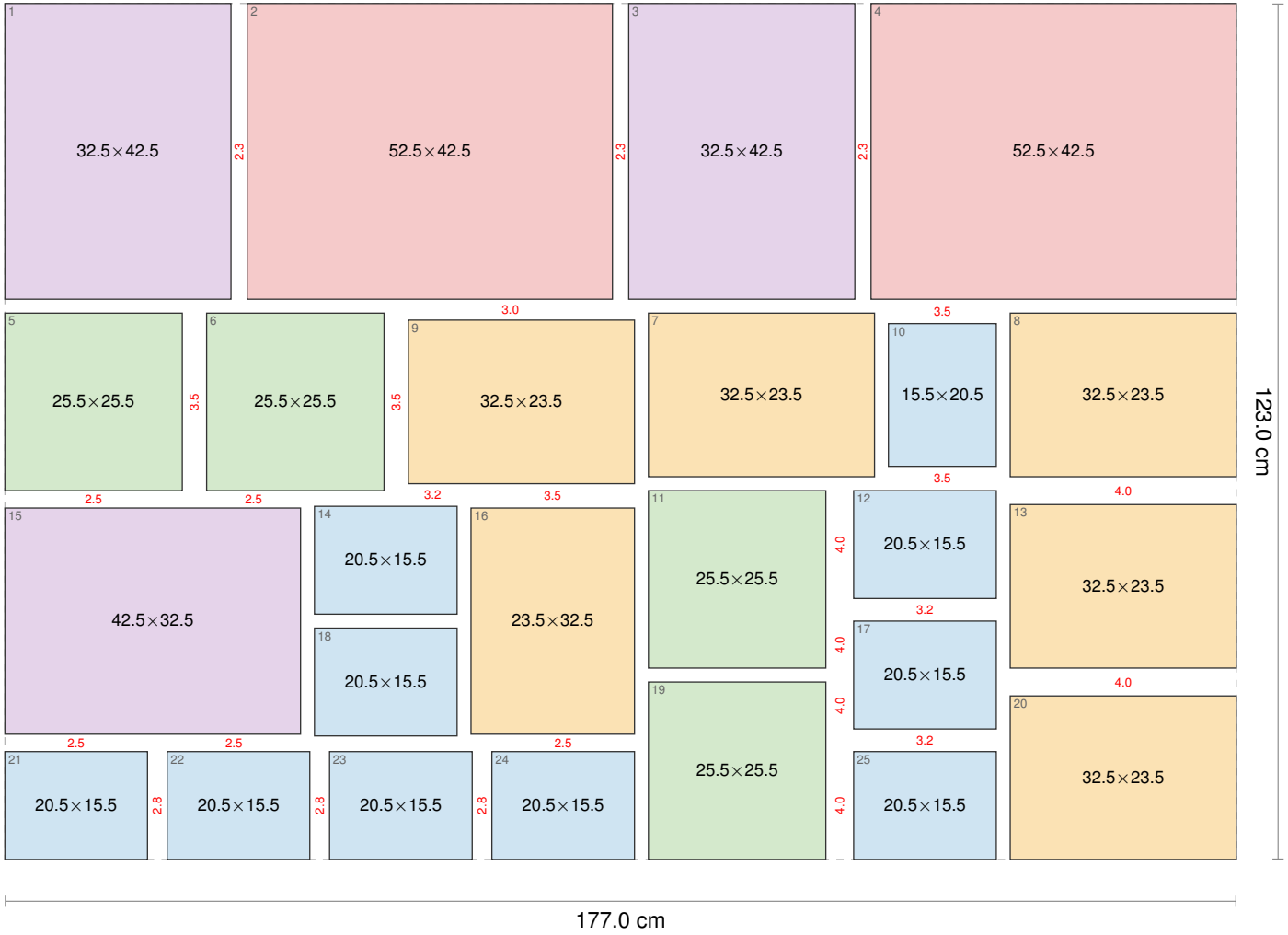
Outer size	172.5 × 116.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 5.5 cm
Max. deviation	3.0 cm
Mean deviation	0.53 cm
Border notches	top 4 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	1
Entropy	0.78
mix / variety	0.81 / 0.78
stagger / lines	0.52 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	32.5 × 23.5	35.0	0.0
3	32.5 × 23.5	72.0	0.0
4	20.5 × 15.5	107.0	0.0
5	42.5 × 52.5	130.0	0.0
6	20.5 × 15.5	107.0	18.5
7	32.5 × 23.5	53.5	27.5
8	15.5 × 20.5	35.0	29.0
9	15.5 × 20.5	89.0	29.0
10	20.5 × 15.5	107.0	37.0
11	32.5 × 42.5	0.0	46.0
12	20.5 × 15.5	35.0	55.0
13	20.5 × 15.5	84.0	55.0
14	20.5 × 15.5	152.0	55.0
15	23.5 × 32.5	58.0	55.5
16	42.5 × 32.5	107.0	55.5
17	20.5 × 15.5	35.0	73.0
18	20.5 × 15.5	84.0	73.0
19	20.5 × 15.5	152.0	73.0
20	25.5 × 25.5	0.0	91.0
21	25.5 × 25.5	63.0	91.0
22	25.5 × 25.5	91.0	91.0
23	25.5 × 25.5	119.0	91.0
24	25.5 × 25.5	147.0	91.0
25	32.5 × 23.5	28.0	93.0

Layout 15 177.0 × 123.0 cm gap 2 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

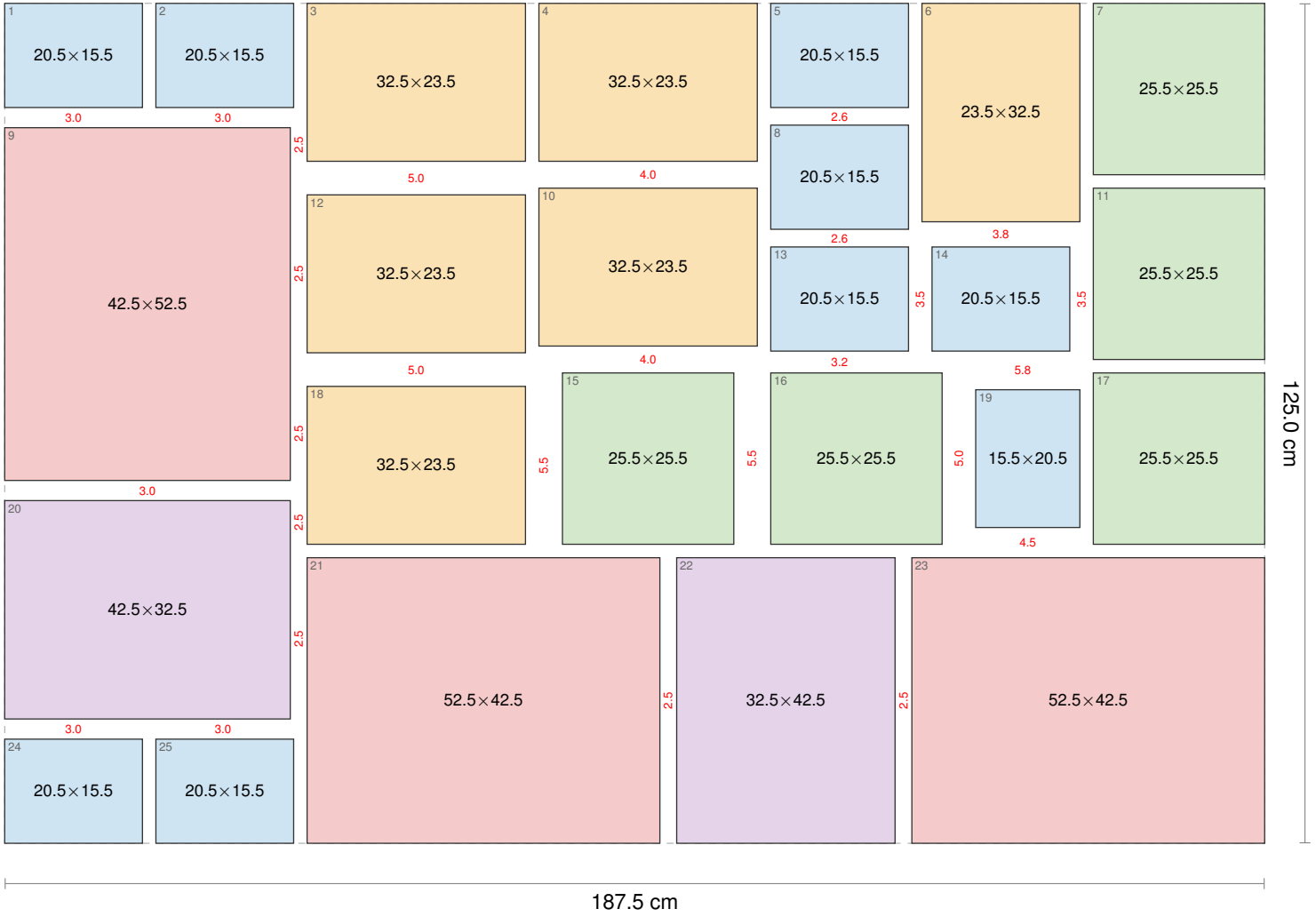
Outer size	177.0 × 123.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.0 cm
Max. deviation	2.0 cm
Mean deviation	0.64 cm
Border notches	bottom 4 cm, right 8 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 90 cm
Entropy	0.79
mix / variety	0.81 / 0.79
stagger / lines	0.54 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	52.5 × 42.5	34.8	0.0
3	32.5 × 42.5	89.7	0.0
4	52.5 × 42.5	124.5	0.0
5	25.5 × 25.5	0.0	44.5
6	25.5 × 25.5	29.0	44.5
7	32.5 × 23.5	92.5	44.5
8	32.5 × 23.5	144.5	44.5
9	32.5 × 23.5	58.0	45.5
10	15.5 × 20.5	127.0	46.0
11	25.5 × 25.5	92.5	70.0
12	20.5 × 15.5	122.0	70.0
13	32.5 × 23.5	144.5	72.0
14	20.5 × 15.5	44.5	72.2
15	42.5 × 32.5	0.0	72.5
16	23.5 × 32.5	67.0	72.5
17	20.5 × 15.5	122.0	88.8
18	20.5 × 15.5	44.5	89.8
19	25.5 × 25.5	92.5	97.5
20	32.5 × 23.5	144.5	99.5
21	20.5 × 15.5	0.0	107.5
22	20.5 × 15.5	23.3	107.5
23	20.5 × 15.5	46.7	107.5
24	20.5 × 15.5	70.0	107.5
25	20.5 × 15.5	122.0	107.5

Layout 16 187.5 × 125.0 cm gap 2 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	187.5 × 125.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 5.8 cm
Max. deviation	3.8 cm
Mean deviation	0.86 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	3, not touching, min. dist. 77 cm
Entropy	0.78
mix / variety	0.84 / 0.78
stagger / lines	0.51 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	1	2
	32.5×42.5	P	1	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	20.5 × 15.5	22.5	0.0
3	32.5 × 23.5	45.0	0.0
4	32.5 × 23.5	79.5	0.0
5	20.5 × 15.5	114.0	0.0
6	23.5 × 32.5	136.5	0.0
7	25.5 × 25.5	162.0	0.0
8	20.5 × 15.5	114.0	18.1
9	42.5 × 52.5	0.0	18.5
10	32.5 × 23.5	79.5	27.5
11	25.5 × 25.5	162.0	27.5
12	32.5 × 23.5	45.0	28.5
13	20.5 × 15.5	114.0	36.2
14	20.5 × 15.5	138.0	36.2
15	25.5 × 25.5	83.0	55.0
16	25.5 × 25.5	114.0	55.0
17	25.5 × 25.5	162.0	55.0
18	32.5 × 23.5	45.0	57.0
19	15.5 × 20.5	144.5	57.5
20	42.5 × 32.5	0.0	74.0
21	52.5 × 42.5	45.0	82.5
22	32.5 × 42.5	100.0	82.5
23	52.5 × 42.5	135.0	82.5
24	20.5 × 15.5	0.0	109.5
25	20.5 × 15.5	22.5	109.5